

Inverting Amplifier with multiple input (Adder Circuit)

(i) Two input $\Rightarrow$

Input current,

$$i = i_1 + i_2$$

output voltage,

$$V_o = -i R_f + V^-$$

$$= -i R_f \quad (V^- = V^+ = 0)$$

$$i_1 = \frac{V_1}{R_1}$$

$$i_2 = \frac{V_2}{R_2}$$

$$\therefore i = i_1 + i_2$$

$$= \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_o = -i R_f = -(i_1 + i_2) R_f$$

$$= -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2}\right) R_f$$

$$= -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)$$

Gain term
for V_1
component

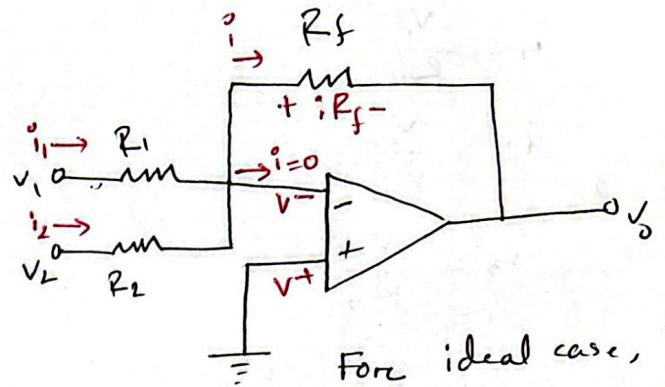
Gain term
for V_2
component

If, $R_1 = R_2 = R$.

$$V_o = -(V_1 + V_2) \frac{R_f}{R}$$

If, $R_f = R = R$

$$\therefore V_o = -(V_1 + V_2)$$



For ideal case,

$$V^- = V^+ = 0V$$

$$R_{id} = \infty, \quad i = 0$$

Weighted Adder
Circuit.

② For three input

$$i_1 = \frac{V_1}{R_1}$$

$$i_2 = \frac{V_2}{R_2}$$

$$i_3 = \frac{V_3}{R_3}$$

$$\therefore i = i_1 + i_2 + i_3$$

$$= \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

$$V_o = -i R_f$$

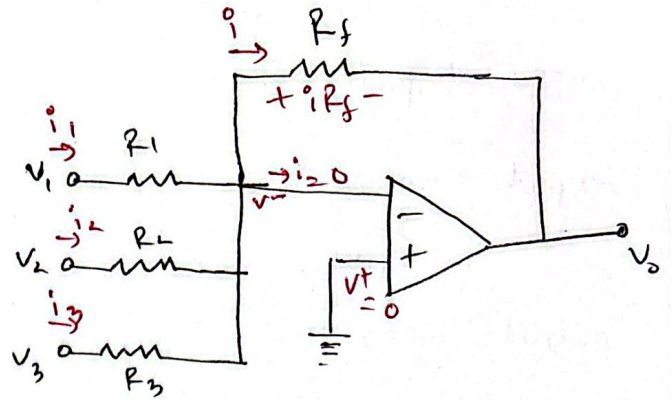
$$= -(i_1 + i_2 + i_3) R_f$$

$$= - \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$= - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

$$= - (V_1 + V_2 + V_3) \frac{R_f}{R} \quad [\text{If } R_1 = R_2 = R_3 = R]$$

$$= - (V_1 + V_2 + V_3) \quad [\text{If } R_f = R]$$



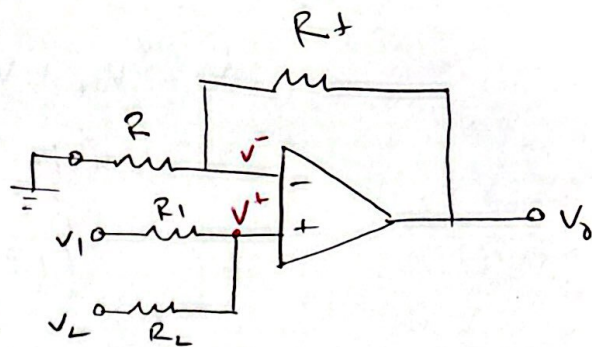
Non-Inverting Op-Amp with multiple input:

Output voltage,

$$V_o = A_{cl} v_i$$

$$= \left(1 + \frac{R_f}{R}\right) v^+$$

$v^+ = ?$ [No virtual ground concept so apply superposition]



Superposition Method

→ Consider only one input at a time

$$v^+ = v_1^+ + v_2^+$$

↙ voltage at non-inverting terminal due to input voltage v_1

→ voltage at non-inverting terminal due to input voltage v_2 .

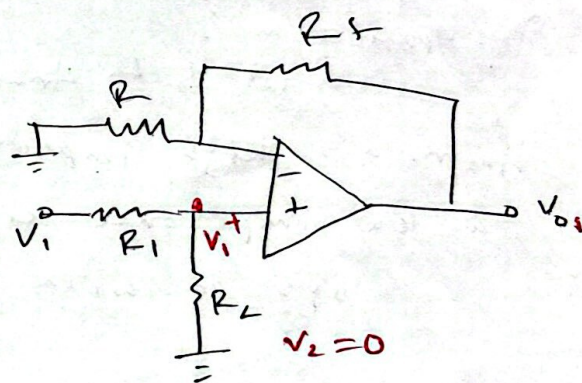
case 1:

v_1 active, v_2 set to zero

$$V_{o1} = A_{cl} v_1^+$$

$$= \left(1 + \frac{R_f}{R}\right) v_1^+$$

$$= \left(1 + \frac{R_f}{R}\right) \left(\frac{R_2}{R_1 + R_2}\right) v_1$$



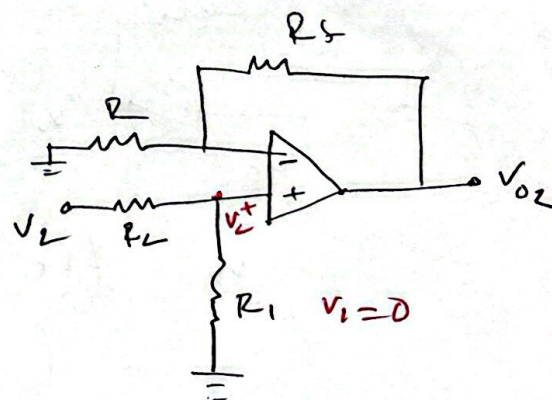
case 2:

v_2 active, v_1 is set to zero

$$V_{o2} = A_{cl} v_2^+$$

$$= \left(1 + \frac{R_f}{R}\right) v_2^+$$

$$= \left(1 + \frac{R_f}{R}\right) \left(\frac{R_1}{R_1 + R_2}\right) v_2$$



Now with both v_1 and v_2 active.

$$V_o = V_{o1} + V_{o2}$$

$$= \left(1 + \frac{R_f}{R}\right) (v_1 + v_2)$$

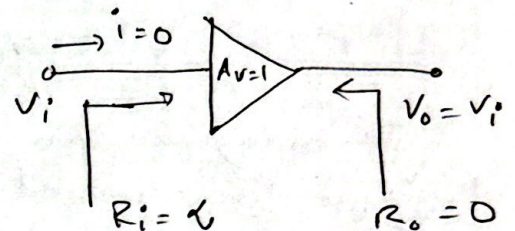
$$= \left(1 + \frac{R_f}{R}\right) \left(\frac{R_L}{R_1 + R_L} v_1 + \frac{R_1}{R_1 + R_L} v_2\right)$$

Voltage Buffer Circuit:

A Voltage Buffer is an amplifier with Unity voltage gain. $v_o = v_i$

Features:

- (1) $A_v = 1$, unity voltage gain
- (2) $R_i = \infty$, Infinite input res
- (3) $R_o = 0$, Zero output res.

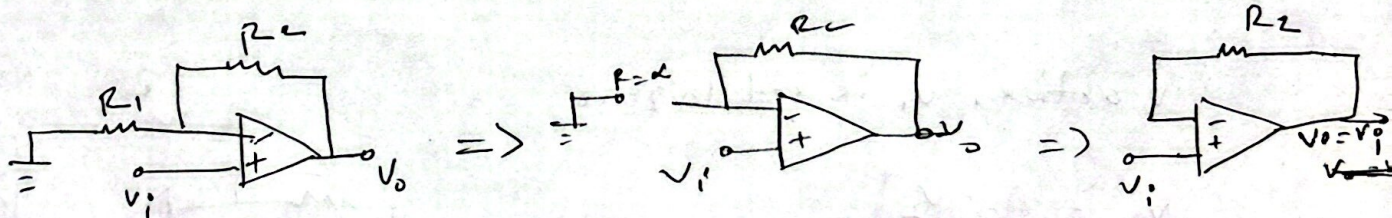


Circuit symbol of voltage Buffer with its main feature.

Application:

=> used to transfer voltage from a high impedance circuit to a low impedance circuit without any loss or with minimum loss.

Non-inverting op-amp $\rightarrow$ Voltage Buffer:



$$v_o = \left(1 + \frac{R_f}{R_i}\right) v_i$$

$$= \left(1 + \frac{R_f}{\infty}\right) v_i$$

$$= v_i$$

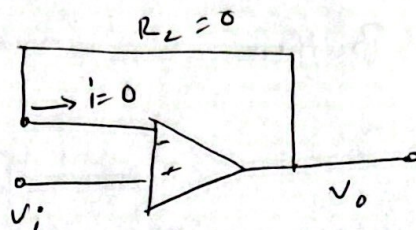
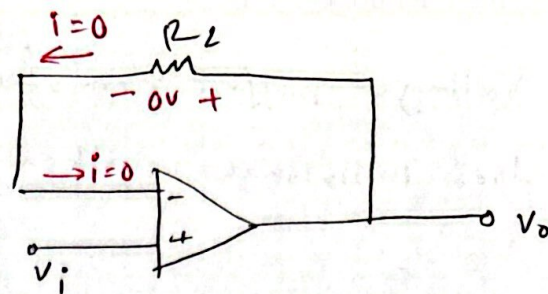
$$\therefore v_o = v_i$$

3

Physical Inspection.

$$V_o = V^- = V_i$$

$$V_o = V^- = V_i$$



Op- Amp based voltage buffers have all the features of an ideal Voltage Buffer.

① $A_v = 1$

② $R_i = \infty$

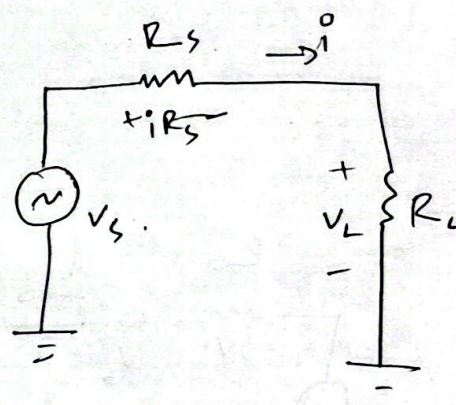
③ $R_o = 0$

Application of voltage Buffer:

$$V_L = \frac{R_L}{R_L + R_S} V_s < V_s$$

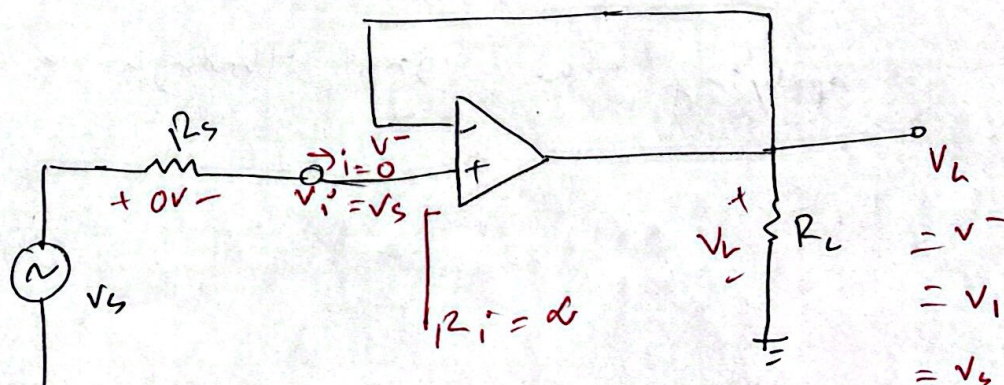
$$V_L = V_s - i R_S$$

$$R_L \downarrow, i \uparrow, i R_S \uparrow, V_L \downarrow$$



The lower is R_L , the lower is V_L .

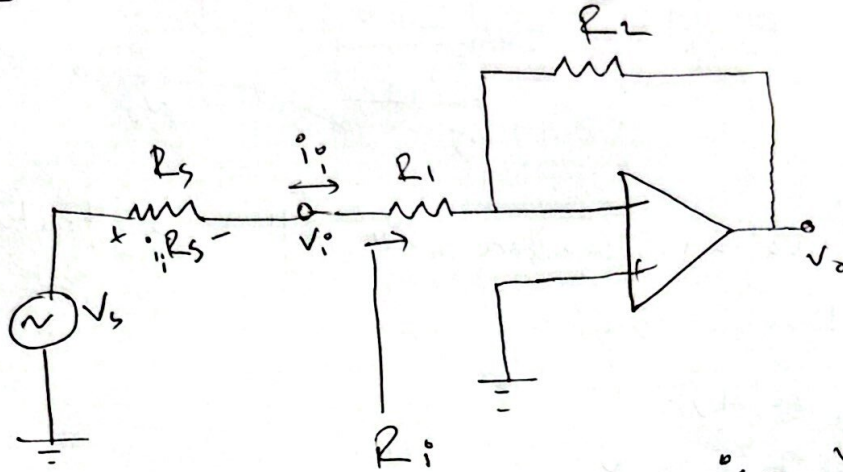
Low resistance amp $\rightarrow$ microphone.



A voltage buffer isolates the output from the input.

As Voltage buffer transfers the signal voltage to the load resistance without any loss -

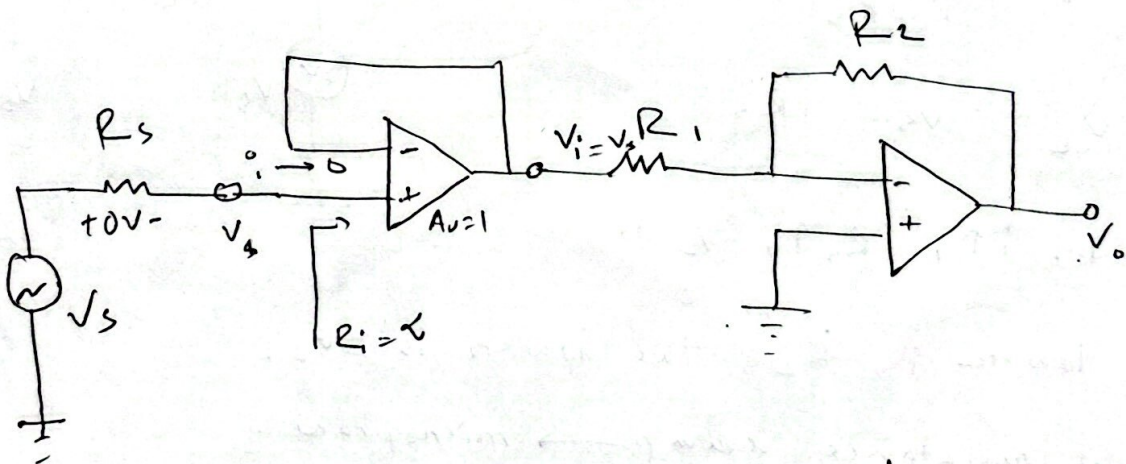
* Voltage Buffer in inverting amp .



$$i_i = \frac{V_i}{R_i} = \frac{V_i}{R_1}, R_i = R_1$$

$$V_i = V_s - i_i R_s < V_s$$

Loss of signal due to finite R_i .



Loss of signal is avoided by applying the signal through a voltage buffer.

Difference Amplifier:

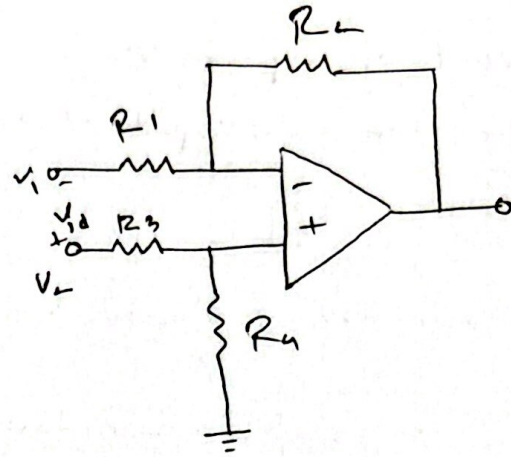
$$V_o = A_d (V_2 - V_1) = A_d V_{id}$$

$V_{id} = V_2 - V_1$, Differential input voltage.

$$A_d = \frac{V_o}{V_2 - V_1} = \frac{R_L}{R_1}$$

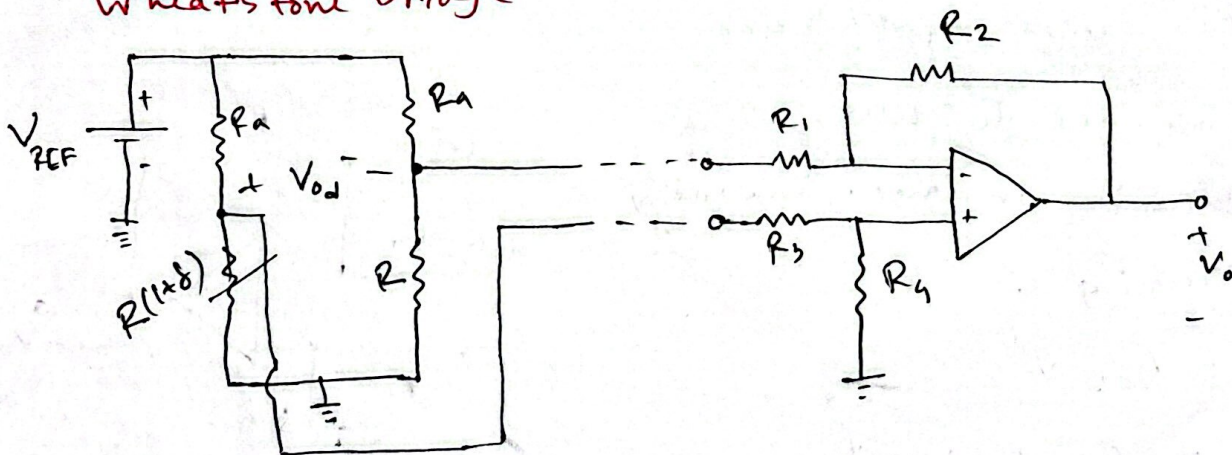
↓
Differential Gain.

provided, $\frac{R_1}{R_L} = \frac{R_3}{R_4}$

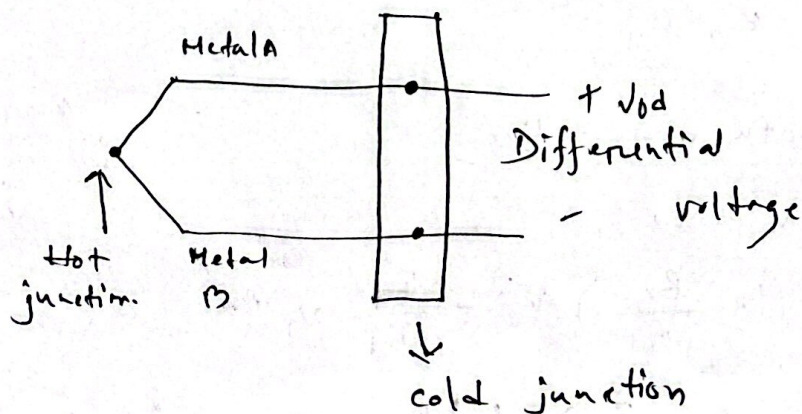


Application:

Wheatstone Bridge



Thermocouple:



* Why not conventional amplifier

⇒ No junction is grounded.

Gain Determination:

Method of Superposition:

→ consider one input at a time.

case 1

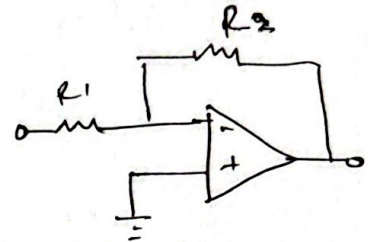
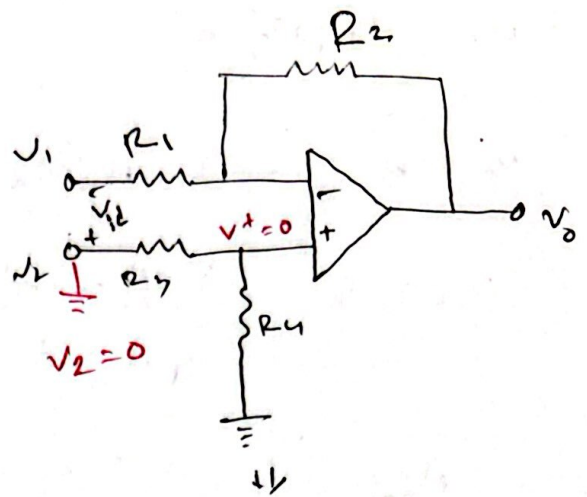
V_1 is active, V_2 is zero

With $V_2 = 0$, amplifier reduces to that of Inverting amplifier configuration.

output voltage due to input,

$$V_{O1} = A_{u1} V_1$$

$$= -\frac{R_2}{R_1} V_1$$



Case 2

V_2 is active V_1 set to zero.

With V_1 is set to zero, amplifier is in non-inverting amp config.

$$V_{O2} = A_{u2} V^+$$

$$= \left(1 + \frac{R_2}{R_1}\right) V^+$$

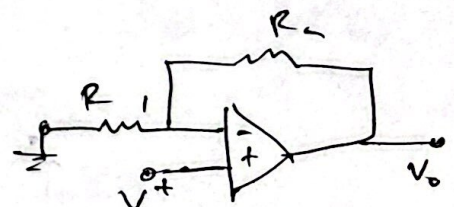
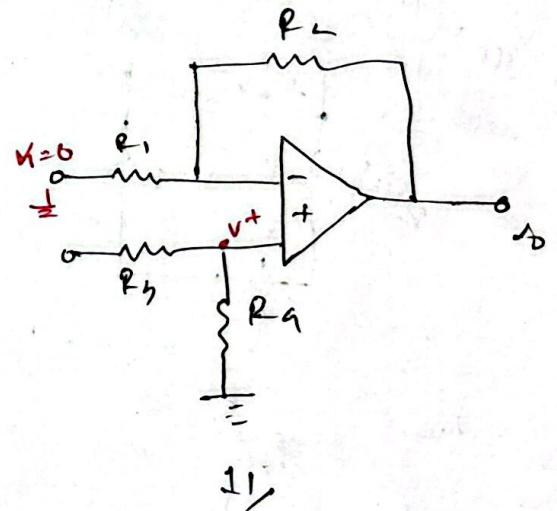
$$V^+ = \frac{R_4}{R_3 + R_4} V_2$$

Case 3 when both are active

$$V_O = V_{O1} + V_{O2}$$

$$= \left(1 + \frac{R_2}{R_1}\right) V^+ - \frac{R_2}{R_1} V_1$$

$$= \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_4}{R_3 + R_4}\right) V_2 - \frac{R_2}{R_1} V_1$$



$$= \frac{R_1 + R_2}{R_1} \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$$

$$\Rightarrow \boxed{v_o = A_d (v_2 - v_1)}$$

Now, af,

$$\frac{R_1 + R_2}{R_1} \frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1} = A_d$$

$$\therefore v_o = A_d (v_2 - v_1)$$

$$A_d = \frac{v_o}{v_2 - v_1} = \frac{R_2}{R_1}$$

$$\rightarrow \frac{R_1}{R_2} \times \frac{R_1 + R_2}{R_1} \times \frac{R_4}{R_3 + R_4} = 1$$

$$\Rightarrow \frac{R_1 + R_2}{R_2} \cdot \frac{R_4}{R_3 + R_4} = 1$$

$$\Rightarrow \frac{R_1 + R_2}{R_2} = \frac{R_3 + R_4}{R_4}$$

$$\Rightarrow 1 + \frac{R_1}{R_2} = 1 + \frac{R_3}{R_4}$$

$$\Rightarrow \boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$$

Input resistance of Difference Amplifier:

$$R_{id} = \frac{V_{id}}{i}$$

$$V_{id} = iR_3 + 0 + iR_1$$

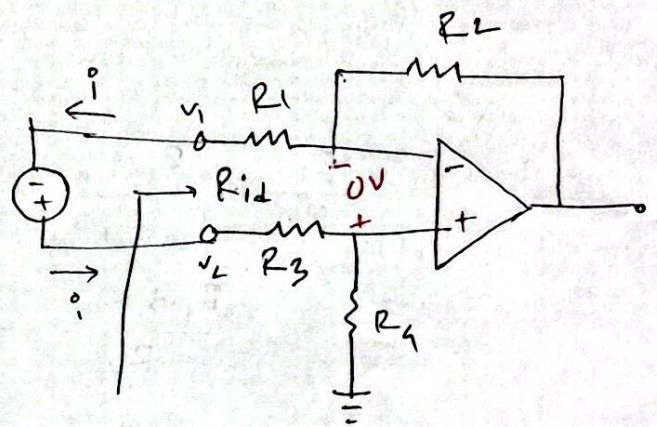
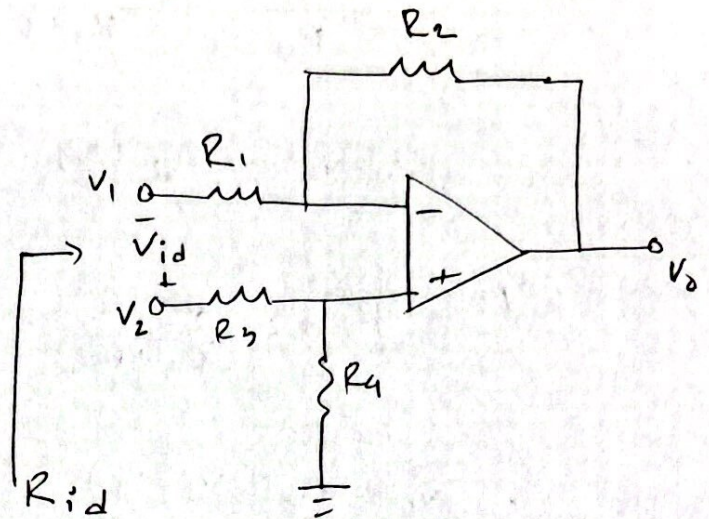
$$= i(R_1 + R_3)$$

$$R_{id} = \frac{V_{id}}{i} = R_1 + R_3$$

$$A_d = \frac{R_2}{R_1}$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Basic Difference Amplifier has a low input Resistance.



$$V_{id} = V_s - i_s R_s$$

$$R_1, R_3 \downarrow \Rightarrow i \uparrow, iR_s \uparrow,$$

$$V_{id} \downarrow, V_o \downarrow,$$

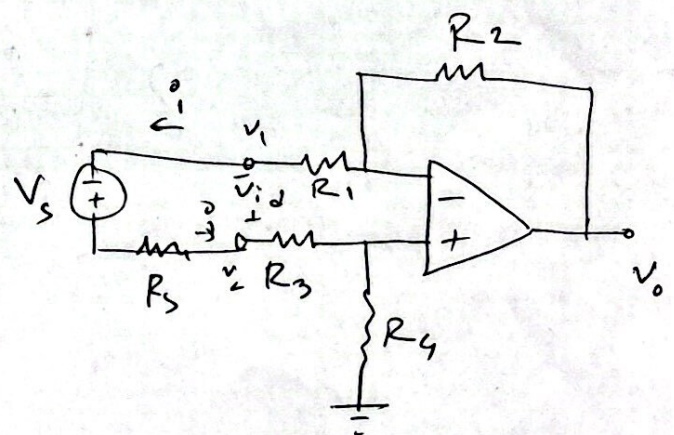
$$V_o = A_d V_{id}$$

Low input Resistance $\rightarrow$

- ① High $i^2 R$ loss
- ② Loss of signal

Limitation:

- ① Low input Resistance
- ② Low Gain
- ③ Difficult to vary gain.



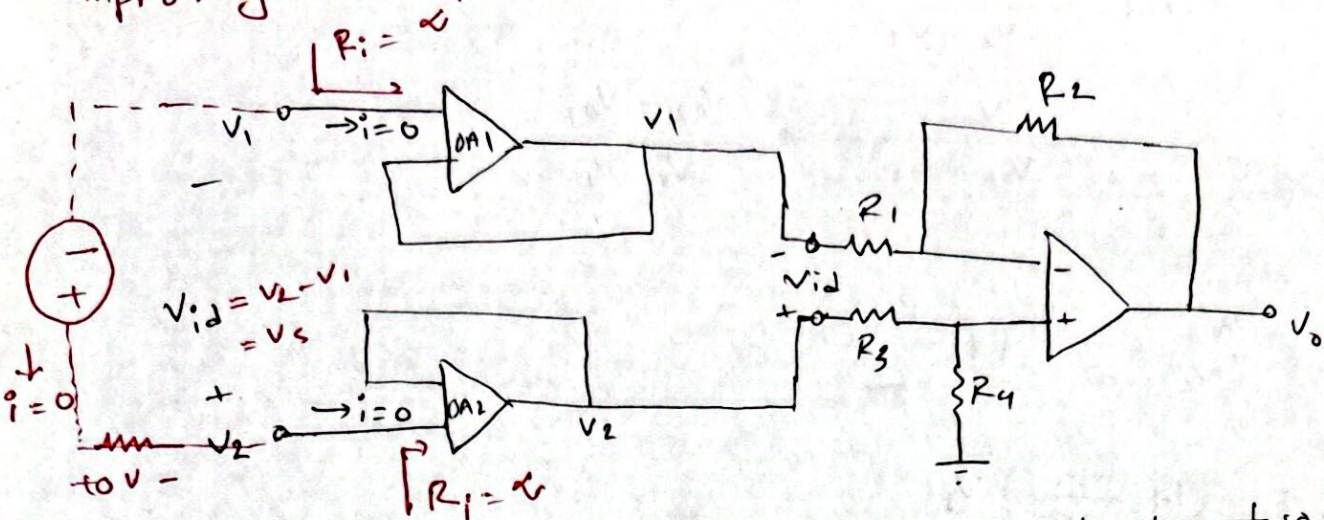
$$A_d = \frac{R_2}{R_1}$$

$$R_{id} = R_1 + R_3$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

7 Instrumentation Amplifier.

Improving the input resistance!



① → introducing two voltage buffers at the two inputs
 → Voltage buffers will at the two inputs will provide ideally infinite input resistance.

② → Signal source is isolated from the amplifier by the voltage buffers.
 → Signal source doesn't supply any current and there is no loss of signal.

Improving the gain!

* Converting the voltage buffers into non-inverting amplifiers by adding two resistors R_f and R .

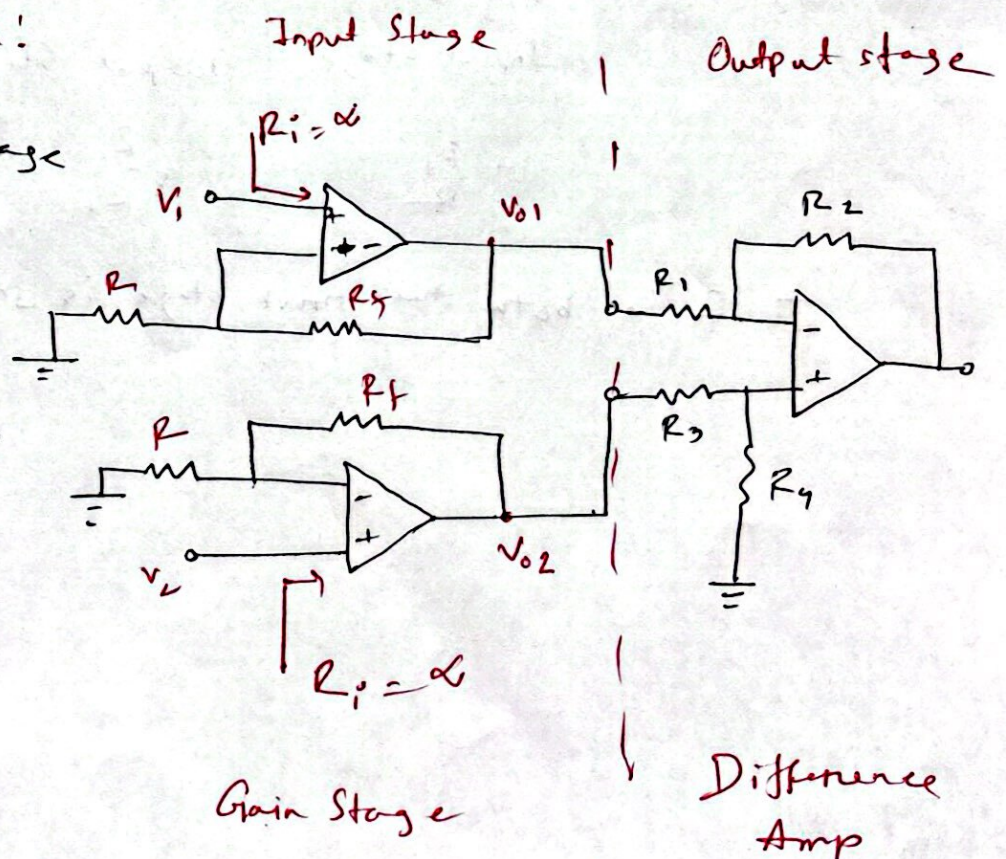


Fig. 2

$$A_d = \frac{V_o}{V_2 - V_1}$$

$$= \frac{V_o}{V_{o2} - V_{o1}} \times \frac{V_{o2} - V_{o1}}{V_2 - V_1}$$

??

$$\frac{V_{o1}}{V_1} = \frac{V_{o2}}{V_2} = 1 + \frac{R_f}{R}$$

$$V_{o1} = \left(1 + \frac{R_f}{R}\right) V_1$$

$$V_{o2} = \left(1 + \frac{R_f}{R}\right) V_2$$

$$V_{o2} - V_{o1} = \left(1 + \frac{R_f}{R}\right) (V_2 - V_1)$$

$$\Rightarrow \frac{V_{o2} - V_{o1}}{V_2 - V_1} = \left(1 + \frac{R_f}{R}\right)$$

$$\therefore A_d = \frac{R_L}{R_1} \left(1 + \frac{R_f}{R}\right)$$

Making the Gain Variable:

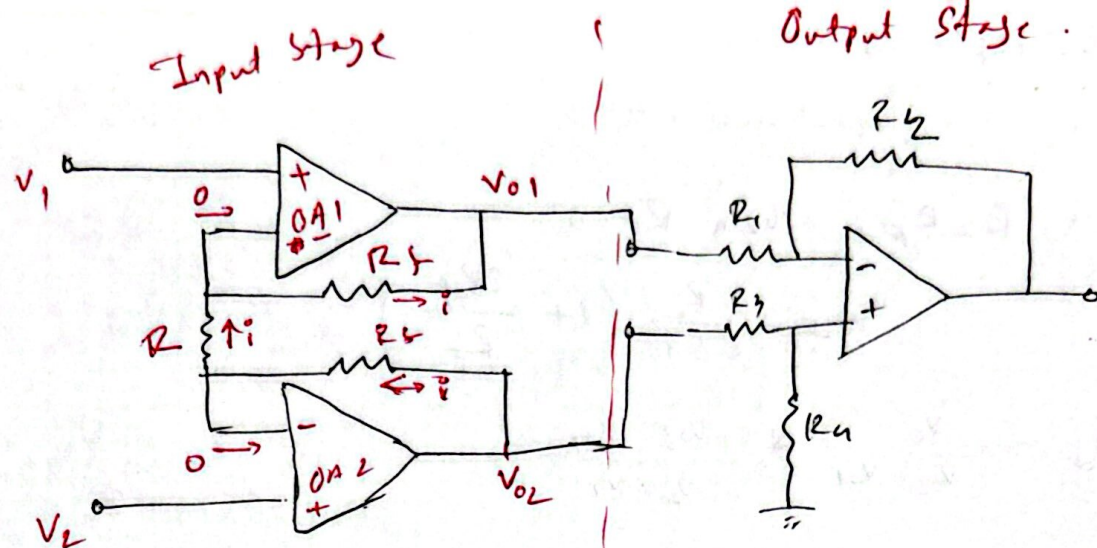
Fig 2

Input stage | Output stage

$$1 + \frac{R_f}{R}$$

$$\& \frac{R_L}{R_1}$$

* R in both the first stage is grounded.



$$A_d = \frac{V_0}{V_{02} - V_{01}} * \frac{V_{02} - V_{01}}{V_2 - V_1}$$

$$V_{02} - V_{01} = i R_f + i R + i R_f$$

$$= i (R + 2R_f)$$

$$= \frac{V_2 - V_1}{R} (R + 2R_f)$$

$$= (V_2 - V_1) \left(1 + \frac{2R_f}{R} \right)$$

$$\frac{V_{02} - V_{01}}{V_2 - V_1} = 1 + \frac{2R_f}{R}$$

$$A_d = \frac{V_0}{V_{02} - V_{01}} * \frac{V_{02} - V_{01}}{V_2 - V_1}$$

$$= \frac{R_2}{R_1} \left(1 + \frac{2R_f}{R} \right)$$

* R can be variable. But it has limited value upto 10k as a potentiometer.

If, $R = 0$, $A_d = \infty$, short circuit which may damage the circuit.

$R = R_F + R_v \rightarrow$ variable resistor
 $\hookrightarrow$ fixed resistor.

34. $R = R_F$ when $R_v = 0$.

$$A_d = \frac{R_L}{R_1} \left(1 + \frac{2R_F}{R_F} \right)$$

$$A_d = \frac{V_o}{V_{oL} - V_{oI}} \times \frac{V_{oL} - V_{oI}}{V_L - V_I}$$

$$= \frac{R_L}{R_1} \left(1 + \frac{2R_F}{R_F + R_v} \right)$$

$$A_{dmin} = \frac{R_L}{R_1} \left(1 + \frac{2R_F}{R_F + R_{vmax}} \right)$$

$$A_{dmax} = \frac{R_L}{R_1} \left(1 + \frac{2R_F}{R_F} \right), R_v = 0$$

Advantages:

• Very high input resistance
(Ideally infinite)

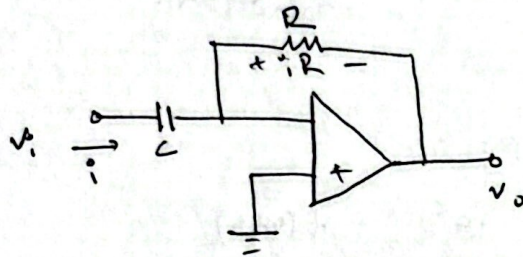
• Very high gain

• Easy to vary gain

$$R_F = 100.2 \Omega$$

$$R_L = 50.050 \text{ K}\Omega$$

Differentiation and Integration Circuit.



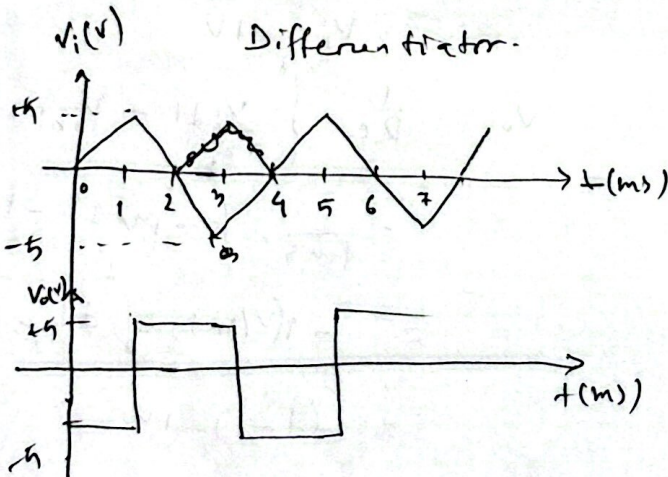
Differentiator Circuit.

$$i_c = C \frac{dv_i}{dt}$$

$$v_o = -iR$$

$$= -RC \frac{dv_i}{dt}$$

Differentiator.

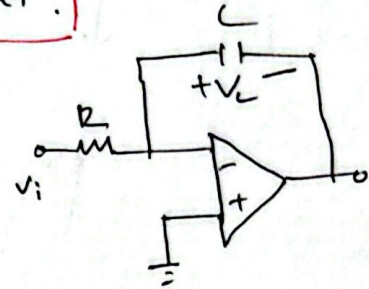


For. $0 < t < 1$, $\frac{dv_i}{dt} = \frac{\Delta v_i}{\Delta t} = \frac{5V}{1ms} = 5V/ms$

$$v_o = -RC \frac{dv_i}{dt} = -5RC = -5V/ms \quad \left[\text{if } R=1K, C=1\mu \right]$$

For. $1 < t < 3$, $\frac{dv_i}{dt} = -\frac{10V}{2ms} = -5V/ms$

$$\therefore v_o = -RC \frac{dv_i}{dt} = 5V/ms$$



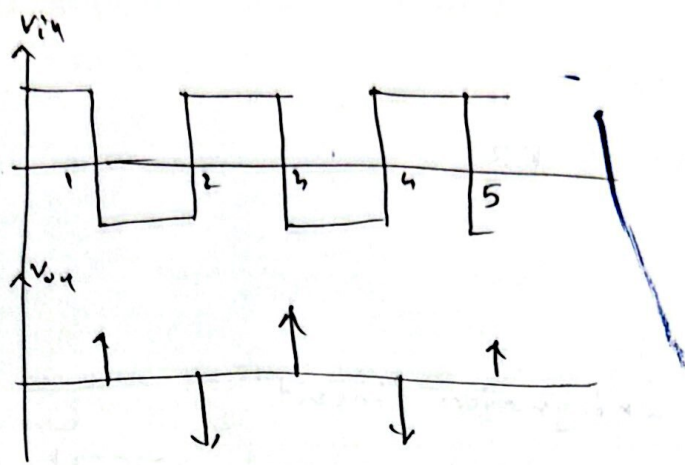
Integrator Circuit

$$v_c = \frac{1}{C} \int i dt$$

$$= \frac{1}{C} \int \frac{v_i}{R} dt$$

$$v_o = -v_c$$

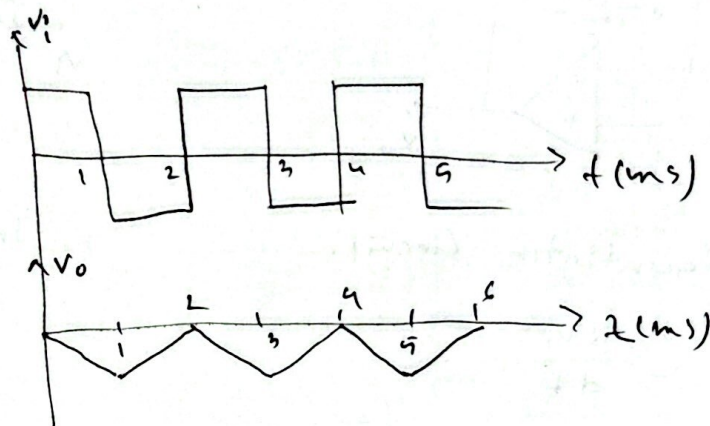
$$= -\frac{1}{RC} \int v_i dt$$



if a diode is connected to output only positive spike.

spikes are impulse function.

Integrator:



For, $0 < t < 1 \text{ ms}$

$$V_i = 1 \text{ V}$$

$$\begin{aligned} V_o &= -\frac{1}{RC} \int_0^t V_i dt + V_{\text{initial}} \\ &= -\frac{1}{1} \int_0^t 1 \text{ V} dt \\ &= -(1 \text{ V/ms})(t \text{ ms}) \end{aligned}$$

$$\therefore V_o = -t \text{ V}$$

$$\text{At, } t=0, V_o=0$$

$$t=1, V_o=-1$$

For, $1 < t < 2$

$$V_i = -1 \text{ V}$$

$$\begin{aligned} V_o &= -\frac{1}{RC} \int_1^t V_i dt + V_{\text{ini}} \\ &= -\frac{1}{1 \text{ ms}} \int_1^t -1 \text{ V} dt - 1 \text{ V} \\ &= -1 (\text{V/ms}) \int_1^t dt - 1 \text{ V} \\ &= (t-1) - 1 \text{ V} \end{aligned}$$

$$V_o = (t-2) \text{ V}$$

$$\text{At, } t=1, V_o = 1-2 = -1 \text{ V}$$

$$\text{At, } t=2, V_o = 2-2 = 0 \text{ V}$$

Weighted Adder Circuit

1. Design an inverting op-amp circuit to form the weighted sum V_o of two inputs V_1 and V_2 . It is required that,

$$V_o = -(V_1 + 5V_2).$$

Choose the values for R_1 , R_2 and R_f so that for a maximum output voltage of 10 V, the current in the feedback resistor will not exceed 1 mA.

Solⁿ:

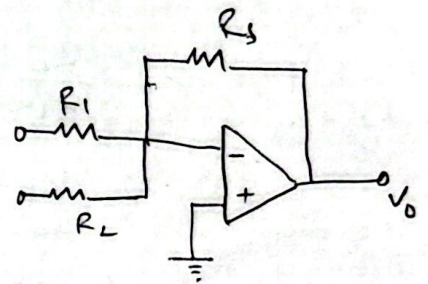
$$V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right) \quad \text{--- (i)}$$

$$V_o = -(V_1 + 5V_2) \quad \text{--- (ii)}$$

Comparing (i) and (ii),

$$\frac{R_f}{R_1} = 1 \Rightarrow R_f = R_1$$

$$\frac{R_f}{R_2} = 5 \Rightarrow R_f = 5R_2 \Rightarrow R_2 = \frac{R_f}{5}$$



To ensure the feedback resistor current is limited to 1 mA max,

$$I_{R_f} = \frac{V_o}{R_f} \leq 1 \text{ mA}$$

$$\Rightarrow R_f \geq \frac{V_o}{I_{R_f}} = \frac{10 \text{ V}}{1 \text{ mA}} = 10 \text{ K}\Omega$$

$$\therefore R_1 = R_f = 10 \text{ K}\Omega$$

$$R_2 = \frac{R_f}{5} = 2 \text{ K}\Omega.$$

Design an adder circuit with voltage expression.

$$V_o = 2V_1 + 3V_2 - 5V_3$$

Choose resistor value not less than $10\text{K}\Omega$.

$$V_o = -(-2V_1 - 3V_2 + 5V_3)$$

$$V_o = - \frac{R_f}{R_i} R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$\therefore \frac{R_f}{R_1} = -2 \Rightarrow R_f = 2R_1 \quad [\text{resistor can't have negative value}]$$

$$\frac{R_f}{R_2} = -3 \Rightarrow R_f = 3R_2$$

$$\frac{R_f}{R_3} = 5 \Rightarrow R_f = 5R_3$$

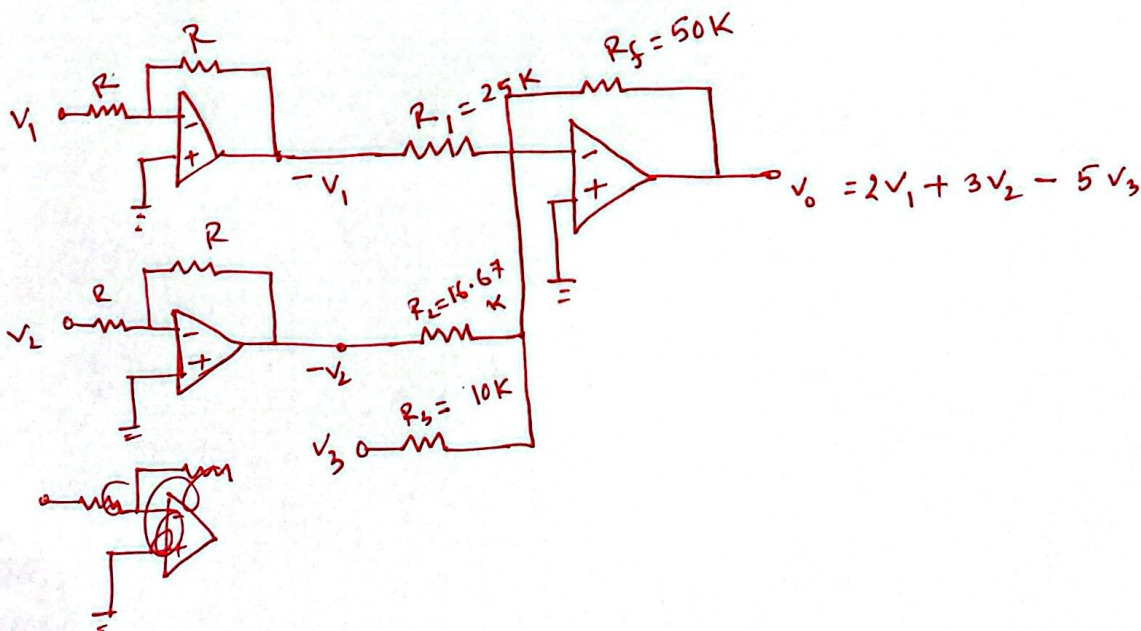
Let,

$$R_3 = 10\text{K}\Omega$$

$$\therefore R_f = 50\text{K}\Omega$$

$$\therefore R_2 = 50/3\text{K}\Omega = 16.67\text{K}\Omega$$

$$\therefore R_1 = 50/2\text{K}\Omega = 25\text{K}\Omega$$



2. Design a noninverting amplifier with a gain of 2. At the maximum output voltage of 10V, the current in the voltage divider is to be 10 mA.

Solⁿ:

$$V_o = V_{in} \left(1 + \frac{R_2}{R_1} \right)$$

It, gain = 2,

$$1 + \frac{R_2}{R_1} = 2$$

$$\Rightarrow \frac{R_2}{R_1} = 1$$

$$\Rightarrow R_1 = R_2$$

Current in the voltage divider,

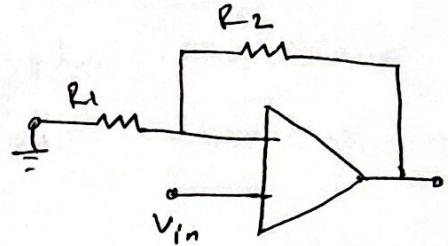
$$I = \frac{V_o}{R_1 + R_2}$$

$$\Rightarrow R_1 + R_2 = \frac{10V}{10mA} = 1M\Omega$$

$$\Rightarrow 2R_1 = 1M\Omega \quad [\because R_1 = R_2]$$

$$\therefore R_1 = 500K\Omega$$

$$\therefore R_2 = 500K\Omega$$



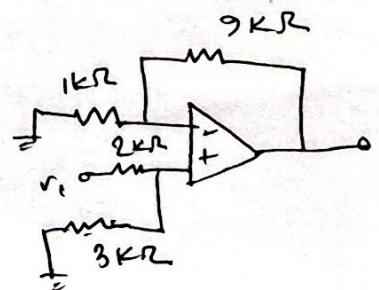
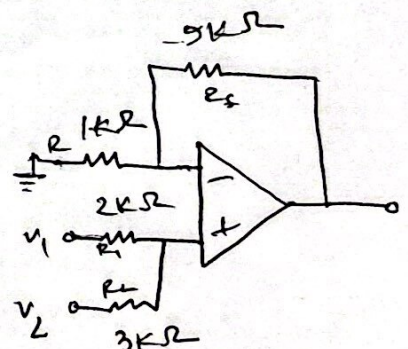
3. Use the superposition principle to find the output voltage of the circuit.

Solⁿ, Case 1: V_2 is set to zero.

V_1 is active.

$$\therefore V_{o1} = \frac{R_2}{R_1 + R_2} V_1$$

$$= \frac{3K\Omega}{5K\Omega} \times V_1 = \frac{3}{5} V_1$$



$$\begin{aligned} \text{Case-2} \\ V_{02} &= \frac{R_1}{R_1 + R_2} V_2 \\ &= \frac{2\text{K}\Omega}{5\text{K}\Omega} V_2 = \frac{2}{5} V_2 \end{aligned}$$

$\therefore$ Case 3: Both active.

The resultant voltage

$$\therefore V_o^+ = V_{01} + V_{02}$$

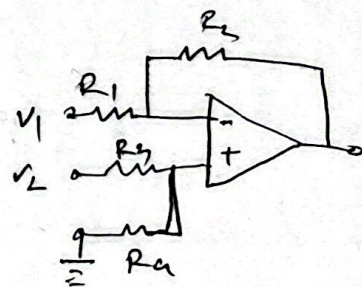
$$= \frac{3}{5} V_1 + \frac{2}{5} V_2$$

$\therefore$ The output voltage,

$$\begin{aligned} V_o &= \left(1 + \frac{R_f}{R}\right) \left(\frac{3}{5} V_1 + \frac{2}{5} V_2\right) \\ &= \left(1 + \frac{9\text{K}\Omega}{1\text{K}\Omega}\right) \left(\frac{3}{5} V_1 + \frac{2}{5} V_2\right) \\ &= 10 \times \left(\frac{3}{5} V_1 + \frac{2}{5} V_2\right) \\ &= 6 V_1 + 4 V_2 \end{aligned}$$

Difference Amplifier:

4. Find the values of the resistor of the circuit so that it behaves as a difference amplifier with a input resistance of 20K and a gain of 10 .



Solⁿ

For difference amplifier:

$$V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

Gain, $\frac{R_2}{R_1} = 10 \quad \therefore R_2 = 10 R_1$

Input resistance,

$$R_{in1} = R_1 + R_3 \quad R_{in2} = R_3$$

$$\therefore R_1 = 20 \text{ K}\Omega$$

$$\therefore R_2 = 10 \times 20 \text{ K}\Omega = 200 \text{ K}\Omega$$

We know,

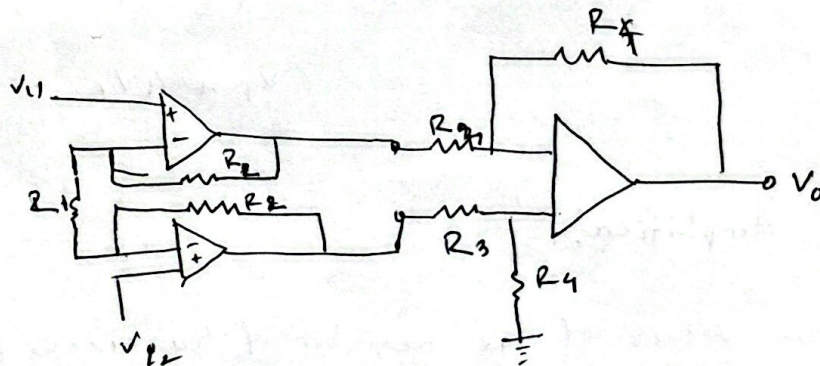
$$\frac{R_2}{R_1} = \frac{R_3}{R_4}$$

$$\therefore R_1 = R_3 = 20 \text{ K}\Omega$$

$$\therefore R_2 = R_4 = 200 \text{ K}\Omega$$

Instrumentation Amp!

5. Design the instrumentation amplifier circuit to provide a gain that can be varied over the range of 2 to 1000 utilizing a $100 \text{ K}\Omega$ variable resistance.



Gain, $1 + \frac{2R_2}{R_F + R_{ov}} = 2 \text{ to } 1000$

$$1 + \frac{2R_2}{R_F} = 1000 \Rightarrow R_F \times 999 = 2R_2$$
$$\Rightarrow R_F = \frac{2R_2}{999}$$

$$1 + \frac{2R_2}{R_F + 100 \text{ K}\Omega} = 2$$

$$R_F = 100.2 \Omega$$

$$R_2 = 50.050 \text{ K}\Omega$$